

1. **Which AI tools did you use? Which were most helpful and why.**

I used two AI tools: **ChatGPT** and **Google Gemini**. ChatGPT was the most helpful for turning rough ideas into a structured “Phase 0” style pitch by organizing the problem, users, features, and potential technologies into a clear proposal format. Gemini was most helpful earlier in the process for fast brainstorming and for generating multiple alternative directions (different app concepts, feature lists, and naming ideas) that I could compare. Overall, ChatGPT was strongest at producing polished writing and an MVP scope, while Gemini was strongest at quickly producing lots of options.

2. **What did AI help with? Where did it fall short?**

AI helped most with idea generation and feature brainstorming. It was useful for:

- suggesting multiple product directions and helping me pick one with a realistic MVP
- listing potential core features and “stretch” features
- proposing a plausible tech stack and basic architecture for a web app
- improving the clarity of the user problem and the value proposition

AI fell short in a few areas:

- It could not confirm anything specific about UT’s real DARS integration, data formats, or permissions, so I had to be careful not to overpromise.
- It sometimes suggested features that are “nice in theory” but complicated in practice (e.g., full official integration, perfect prerequisite planning, real-time university data).
- It did not know class-specific expectations (scope, grading rubric details, team size constraints) unless I provided them.

3. **Would you use it again for this kind of work? Why or why not?**

Yes, I would use AI again for early-stage ideation and pitching. It speeds up the “blank page” stage by producing multiple directions quickly, and it helps refine wording into a concise pitch. However, I would still rely on human judgment for feasibility, constraints, and anything involving real university systems (privacy, authentication, access, and data accuracy). In other words, it’s best as a brainstorming partner and writing assistant, not as the final authority on what is implementable.

4. **What prompts did you provide? Give a summary of the ideas generated.**
Prompts I used (summary):

- “Give me several group project ideas for a CS class that are user-centered and demo-able.”

- “Generate project ideas that would be useful for university students, especially around planning and productivity.”
- “Suggest an app concept that modernizes degree progress tracking and helps students understand what classes they still need.”
- “What features should an MVP include for a DARS-like degree audit dashboard? What tech stack would fit?”
- “Help turn this idea into a Phase 0 proposal with problem, features, users, and technologies.”

Ideas generated (summary):

Across the tools, the AI suggested several categories of projects such as a habit/study planner, a campus study-spot finder, a group project contribution tracker, and a course planning/scheduling tool. After comparing options, I chose a UTK-focused degree progress and course tracking UI (“sits over DARS”) because it solves a real pain point: students struggle to interpret audits and translate them into an actionable semester plan. The final selected concept became a modern requirement checklist/tree, “what counts” course matching, alerts for missing prerequisites, and a simple semester planner.

Final Proposal:

Project Summary

VolTrack is an interactive web application designed to simplify academic progress tracking and schedule planning for students at the University of Tennessee, Knoxville. By converting static degree audit documents into a dynamic, structured visual interface, VolTrack helps students clearly understand their remaining degree requirements and build optimal course schedules for upcoming terms.

Problem Statement

The current Degree Audit Reporting System (DARS) presents academic evaluations as long, text-heavy documents that are difficult to interpret quickly. Students often exit academic advising sessions or registration portals uncertain about remaining requirements, eligible course options, or prerequisite sequences. This lack of clarity leads to missed prerequisites, delays in graduation timelines, and inefficient advising appointments. VolTrack resolves this issue by parsing raw audit data into an actionable, user-friendly visual roadmap.

Major Features

VolTrack organizes academic progress into an intuitive requirement hierarchy, categorizing degree components like General Education, Major Core, Technical Electives, and Minors with clear status indicators for completed, in-progress, and unfulfilled coursework. For any unmet requirement, an integrated "What Counts?" panel lists eligible course options alongside their prerequisites and credit hours. The system includes an intelligent schedule planner that generates recommended course loads based on unfulfilled requirements and active prerequisite chains. It also automatically flags critical constraint conflicts, such as broken prerequisite dependencies or catalog-year discrepancies. Finally, a Built-in "What-If" mode allows students to simulate major or minor changes, while an exportable summary view provides a clean document students can bring directly to advising meetings.

Technology Stack

The frontend is built with React, TypeScript, and Tailwind CSS to deliver a fast, responsive user interface and clean component architecture. The backend utilizes Next.js API routes (or Node.js with Express) to manage application logic and server endpoints. Data storage relies on PostgreSQL to save user profiles, custom schedule plans, and parsed audit states. To handle audit processing, a dedicated server-side ingestion pipeline parses uploaded PDF documents or plain text files, extracting raw audit data and validating it against a structured JSON schema using Zod.

Target Audience and Value Proposition

The primary users are UTK undergraduate students who need an intuitive tool to eliminate the guesswork from degree planning, identify missing prerequisites early, and streamline course registration. The secondary target audience includes academic advisors, who benefit from a clean, standardized overview of student progress. This allows advisors to spend less time interpreting dense audit text and more time focusing on high-level academic and career guidance.